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Efficient feature selection and fusion for real-time beef cattle detection

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ABSTRACT

Real-time and accurate beef cattle detection is essential for effective livestock management. Traditional manual observation methods are labor-intensive and inefficient. Recent studies have shown that deep learning has significantly improved beef cattle detection accuracy. However, achieving robust beef cattle detection remains challenging due to single farming scenarios, occlusions, and dense cattle groups. As an effective solution, this paper proposes a novel method for efficient feature selection and fusion for real-time beef cattle detection (EFSF-RBCD). Specifically, we begin by developing a feature extraction network based on multipath cooperative and poly kernel inception (MPCPKI), which is designed to optimize the feature extraction capabilities. The network includes an efficient P4 feature-layer selection module based on the multipath cooperative gating mechanism (EP4MCGM), which integrates low-level features from shallow layers and enhances fine detail detection. Additionally, the P5 feature layer selection module, based on the cross-stage partial poly kernel inception network (CSPPKINetP5), enables efficient target feature extraction while reducing the computational load. Furthermore, we propose a frequency-domain context feature fusion network (FDCFN), a novel framework that integrates the frequency-domain branch (FDB) and context feature fusion branch (CFFB) to capture local and global contextual information better. Additionally, to enhance detection accuracy, a novel bounding box regression loss function, SloU, was introduced, which improves bounding box position and size estimation by incorporating orientation information between the ground truth and predicted boxes. Experimental results show that EFSF-RBCD achieves an AP@0.5 of 90.3% and an AP@0.5–0.95 of 59.6%, with 26.4M parameters, a computational cost of 50.8 GFLOPs, and a processing speed of 100.3 FPS. The proposed method outperforms existing state-of-the-art methods in terms of AP@0.5 and AP@0.5–0.95 while maintaining a low parameter count and computational load. Additionally, it demonstrated competitive performance in terms of FPS. This study provides a new approach for beef cattle detection in complex environments and lays a theoretical foundation for the development of technologies related to smart-farm deployment.

1. Introduction

Modern livestock farming is increasingly adopting green, efficient, and precise farming supported by information and intelligent technologies (Zheng et al., 2023; Yang et al., 2022b; Liu et al., 2021a; Kumar et al., 2018). Due to its rich nutritional value and growing consumer demand, beef has a broad market demand, and its industry chain is constantly developing and enriching. Detecting beef cattle facilitates large-scale livestock management and has great significance for the intelligent perception of individual cattle behavior analysis and health status (Mar et al., 2023).

Various object detection algorithms have been developed to address the challenges of recognizing and locating objects in complex environments. Early object detection algorithms relied on threshold segmentation (Yin et al., 2012; Amiriebrahimabadi et al., 2024; Feng et al., 2023), frame difference methods (Kristensen and Cornou, 2011), and other techniques to detect animals. For example, Nasirahmadi et al. (2016) employed background subtraction and Otsu's method to convert grayscale images into binary images, effectively segmenting the foreground pixels (pigs) from the background (pens) in group

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environments. Similarly, [Lu et al. \(2016\)](#) used an image-splitting algorithm based on shape characteristics and ellipse fitting to classify resting piglets into different states. However, these traditional detection algorithms face several challenges, including sensitivity to noise and lighting changes, reliance on manually designed features, and difficulties in handling occlusions and dynamic backgrounds, which hinder their application in complex real-world scenarios.

To overcome these limitations, object detection algorithms incorporate mathematical statistics and probability theory ([McLachlan and Rathnayake, 2014](#); [Goyal and Singhai, 2018](#); [Adams and Beling, 2019](#); [Fatemi et al., 2018](#); [Pisner and Schnyer, 2020](#)) to establish dynamic background models. For example, [Yiyang et al. \(2017\)](#) proposed an improved adaptive Gaussian Mixture Model algorithm for individual pig object detection that allows the model to adapt to environmental changes through the adaptive adjustment of model parameters. Despite these advancements, such methods still typically rely on manually selected features, such as color, texture, and shape, which may not fully capture complex patterns in images ([Tian et al., 2024](#); [Sirisha et al., 2023](#)).

Recent advancements in deep learning have led to object detection algorithms ([Girshick et al., 2014](#); [Ren et al., 2016](#); [Redmon and Farhadi, 2017](#); [Carion et al., 2020](#); [Zou et al., 2023](#); [Kamath and Renuka, 2023](#); [Amjoud and Amrouch, 2023](#); [Zhao et al., 2024b](#); [Badgujar et al., 2024](#)) that offer improved solutions for livestock recognition and localization in complex scenarios. Three major model families have emerged: R-CNN-based, YOLO-based, and transformer-based algorithms.

The R-CNN series ([Girshick et al., 2014](#)) has played a foundational role in object detection algorithms. The original R-CNN used selective search to generate region proposals; subsequent versions such as Fast R-CNN ([Girshick, 2015](#)) and Faster R-CNN ([Ren et al., 2016](#)) have introduced significant improvements. Fast R-CNN integrates region proposal generation and classification into a single network, improving both speed and accuracy. Faster R-CNN further improves performance by introducing a region proposal network (RPN) that shares convolutional layers with the detection network, making region proposal generation nearly cost-free. Building on this foundation, [Noe et al. \(2022\)](#) extended the R-CNN framework by implementing Mask R-CNN for precise cattle instance segmentation, achieving high detection accuracy in identifying estrus behaviors of cattle in complex outdoor environments. However, despite setting high benchmarks for detection accuracy, its computational complexity and relatively slower inference times pose challenges for real-time applications, particularly in dynamic and resource-constrained environments such as cattle farms. Differently from the R-CNN series, our method not only improves detection accuracy but also optimizes computational complexity and inference speed, ensuring real-time performance in dynamic and resource-constrained environments like cattle farms.

The You Only Look Once (YOLO) series introduced a paradigm shift in real-time object detection. YOLO models predict bounding boxes and class probabilities directly from full images in a single evaluation, significantly improving detection speed. The YOLO series has evolved to improve accuracy, efficiency, and usability from YOLOv1 ([Redmon et al., 2016](#)) to YOLOv5 ([Jocher, 2020](#)) and YOLOv8 ([Jocher et al., 2023](#)). For instance, [Zin et al. \(2022\)](#) combined YOLOv3-based systems with edge detection for cattle behavior classification, achieving high accuracy in behavior transition recognition. [Qiao et al. \(2023\)](#) developed the YOLOv5-ASFF model for detecting specific cattle body parts, while ([Yang et al., 2023](#)) enhanced YOLOv8 for faster cattle detection by integrating deformable convolutions and coordinate attention mechanisms. More recently, [Mon et al. \(2024\)](#) improved this approach by integrating YOLOv8 with multi-environment tracking, thereby further improving the robustness of real-time cattle identification and tracking. Although YOLO models are well-suited for real-time applications because of their high inference speeds, they struggle with detecting small objects and handling occlusions, which are common challenges in beef cattle detection. Unlike YOLO models, our method

incorporates an MPCPKI-based feature extraction network, enhanced by the EP4MCGM module for fine-detail detection and the CSPPKINetP5 module for efficient target feature extraction. Additionally, the frequency-domain context feature fusion network (FDCFN) captures both local and global contextual information, whereas the SIOU loss function further improves detection accuracy. This comprehensive approach enhances robustness in addressing these challenges.

Transformer-based architectures, such as DETR ([Carion et al., 2020](#)), Deformable DETR ([Zhu et al., 2021](#)), Group DETR ([Chen et al., 2023a](#)), and DAB-DETR ([Liu et al., 2022](#)), have also been applied to object detection tasks. Initially developed for natural language processing, transformers excel in capturing global dependencies and contexts in images using self-attention mechanisms. DETR eliminates the need for traditional components such as RPNs by directly predicting bounding boxes and class labels while leveraging a set-based global loss to address duplicate detections. Although Transformer-based models achieve high accuracy and perform well in complex scenes, their high computational demands often make it difficult to guarantee real-time performance, thereby limiting their practical applications. In contrast, our method is optimized for real-time cattle detection, addressing challenges such as occlusions, environmental variations, and dense cattle distributions in farm settings. Ensuring high detection accuracy while maintaining real-time performance provides a more reliable foundation for practical deployment.

Despite advancements in deep-learning-based object detection algorithms, most focus on detecting individual animals or sparsely distributed groups ([Ardö et al., 2018](#); [Xiong et al., 2024](#); [Mar et al., 2023](#); [Pu and Liu, 2024](#)). However, real farm environments introduce additional challenges, including high animal density, occlusions, and complex backgrounds, which may affect the accuracy and robustness of detection systems ([Ariza-Sentís et al., 2024](#)). Achieving high-precision real-time detection in dynamic environments, complex backgrounds, and dense occlusions remains a significant research problem.

In this study, we compiled a real-world beef cattle object detection dataset, Beef-Cattle. This dataset was collected across diverse scenarios and includes cattle images under various challenging conditions, such as occlusions, shadows, and background clutter. This dataset better reflects real cattle farm conditions, where cattle are raised in groups and are not always centered in images. Using this dataset, we propose a novel method for efficient feature selection and fusion for real-time beef cattle detection (EFSF-RBCD). We first developed a Feature Extraction Network based on multipath cooperative and poly kernel inception (MPCPKI) to optimize the feature extraction capabilities. To integrate various low-level features extracted by the shallow network for fine detail, the network incorporates a P4 feature layer selection module based on the multipath cooperative gating mechanism (EP4MCGM), enhancing fine-detail detection. Next, to efficiently extract target features while reducing the computational load, we introduce the P5 feature layer selection module based on the cross-stage partial poly kernel inception network (CSPPKINetP5), optimizing both the performance and computational efficiency. Additionally, to better capture both local and global contextual information, we propose an FDCFN, which integrates the frequency-domain branch (FDB) and context feature fusion branch (CFFB). Finally, to further improve the detection accuracy, we introduce the SIOU bounding box regression loss function, which enhances the learning of the box position and size by incorporating orientation information.

The key contributions of this study are summarized as follows: (1) We propose a novel method for EFSF-RBCD that significantly enhances the accuracy and robustness of cattle detection in various challenging scenarios. (2) We developed two efficient feature layer selection modules: a P4 module based on a multipath cooperative gating mechanism that integrates various low-level features extracted by the shallow network, improving fine-detail detection, and a P5 layer selection module based on the cross-stage partial poly kernel inception (PKI) network, which enables efficient extraction of target features while

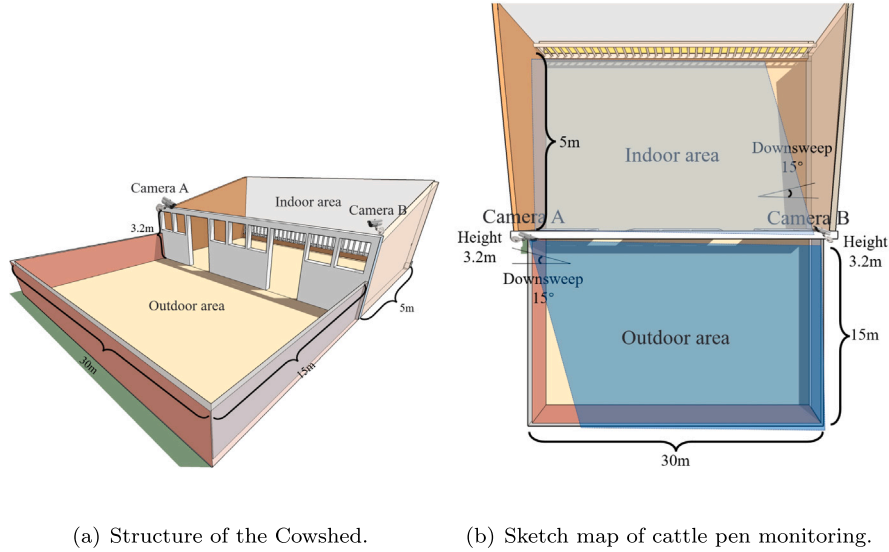


Fig. 1. Sketch map of the interior and exterior structure of a cattle pen.

reducing computational overhead. This combined approach improves the feature extraction efficiency and overall detection performance. (3) The FDCFN, a novel framework that integrates FDB and CFFB to capture both local and global contextual information effectively. Experimental results on a newly compiled beef cattle dataset demonstrate the competitive performance of the proposed method in the cattle object detection task.

2. Material and methods

2.1. Dataset

Experimental video data were collected from a beef cattle farm operated by an ecological breeding company in Yinchuan City, China. These videos were recorded using internal surveillance equipment on the farm that captured the daily activities of the beef cattle. The recorded subjects were adult Angus and Simmental cattle. The test videos were recorded between July 21st and 29, 2019, with sampling times ranging from 5:00 AM to 9:00 PM. The structure of a cattle shed at the farm is shown in Fig. 1(a). The outdoor area measured 30 m × 15 m (length × width), while the indoor area measured 30 m × 5 m (length × width). The video surveillance details of the beef cattle are shown in Fig. 1(b). The installed camera model (*HIKVISION DS-IPC-T12H-1a*) was mounted on the ceiling along the shorter side of the rectangular shed, 3.2 m above the ground, tilted downwards at a 15-degree angle, to cover most areas of cattle activity.

Video segments from four different breeding scenarios were selected. Each video averaged 35 min and was recorded in MP4 format with a resolution of 1920 × 1080 pixels and a frame rate of 24 fps. DarkLabel was used to annotate the selected beef cattle videos. Finally, we constructed a beef cattle detection dataset, Beef-Cattle, containing 11,936 images as the training set, 1492 images as the validation set, and 1492 images as the test set. The dataset was split into training (80%), validation (10%), and test (10%) sets.

In the collected video images, several factors affected the beef cattle detection (Fig. 2:): (1) **Occlusion**. Cattle obstructing each other, as well as barriers like roofs, fences, and dust disturbances caused by running, make detection more challenging. (2) **Light Intensity**: Extended sampling periods result in varying lighting conditions, with shadows from covered sheds and differing color distributions under various lighting conditions affecting the detection. (3) **Scale Variation**: Fixed cameras positioned at the top corners result in size distortion, with nearby cattle appearing larger and distant ones smaller, requiring the

network to extract multiscale features. (4) **Background Clutter**: The muddy ground color closely matches that of cattle, with the background area occupying a much larger portion than the target area, increasing detection complexity.

2.2. EFSF-RBCD

2.2.1. Overview

Fig. 3 illustrates the overall structure of the EFSF-RBCD model, which comprises a feature extraction network based on multipath cooperative and poly kernel inception (MPCPKI), an FDCFN, an IoU-aware query selection mechanism, and a decoder with auxiliary prediction heads. Initially, the input image is processed through the MPCPKI to generate feature maps at three levels: $P_3 \in \mathbb{R}^{128 \times \frac{H}{8} \times \frac{W}{8}}$, $P_4 \in \mathbb{R}^{256 \times \frac{H}{16} \times \frac{W}{16}}$, and $P_5 \in \mathbb{R}^{512 \times \frac{H}{32} \times \frac{W}{32}}$, where H and W represent the height and width of the input image, respectively. The P_5 feature map is first sent to the frequency-domain branch of the FDCFN, where the high- and low-frequency features interact, producing an F_5 feature map. Next, the P_3 , P_4 , and F_5 feature maps pass through the Context Feature Fusion branch of the FDCFN for cross-scale feature fusion and concatenation to produce a feature output called memory. This memory is then fed into the IoU-aware Query Selection to generate the top-K features by computing the regression and classification losses between the predicted bounding boxes corresponding to each feature in the memory and the true bounding boxes. Finally, the top-K features are combined with the memory features as inputs to the Decoder&Head to predict both the class and coordinates of the detection boxes.

2.2.2. Feature extraction network (MPCPKI)

Table 1 presents the details of the proposed MPCPKI feature extraction network. Unlike previous single-path feature extraction networks, MPCPKI integrates multipath cooperative processing and poly kernel inception, allowing it to achieve selective feature enhancement while minimizing redundant computations. This integration enables a more efficient representation learning, making it suitable for complex visual tasks.

To enhance the selection of informative low-level features from the shallow network, we introduce an efficient P_4 feature-layer selection module based on a multipath cooperative gating mechanism (EP4MCGM), as illustrated in Fig. 4. Given an input feature map $P_3 \in \mathbb{R}^{C \times H \times W}$, where C is the number of channels, and H and W represent 1/8 of the height and width of the input image, respectively, EP4MCGM, based on the multipath cooperative gating mechanism

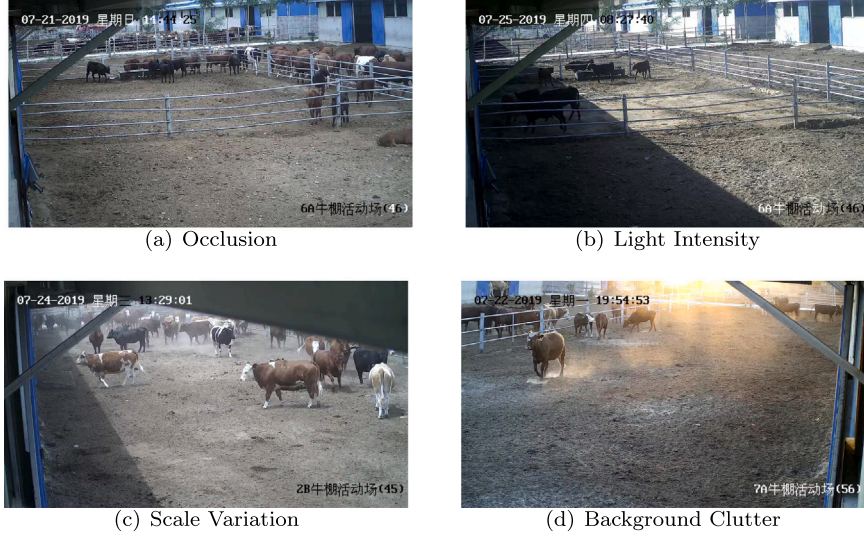


Fig. 2. Sample images of complex scenes.

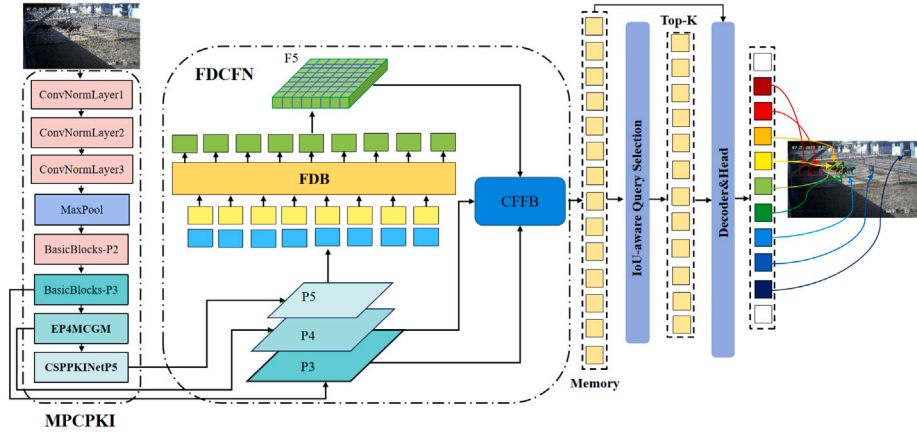


Fig. 3. Architecture of the proposed EFSF-RBCD.

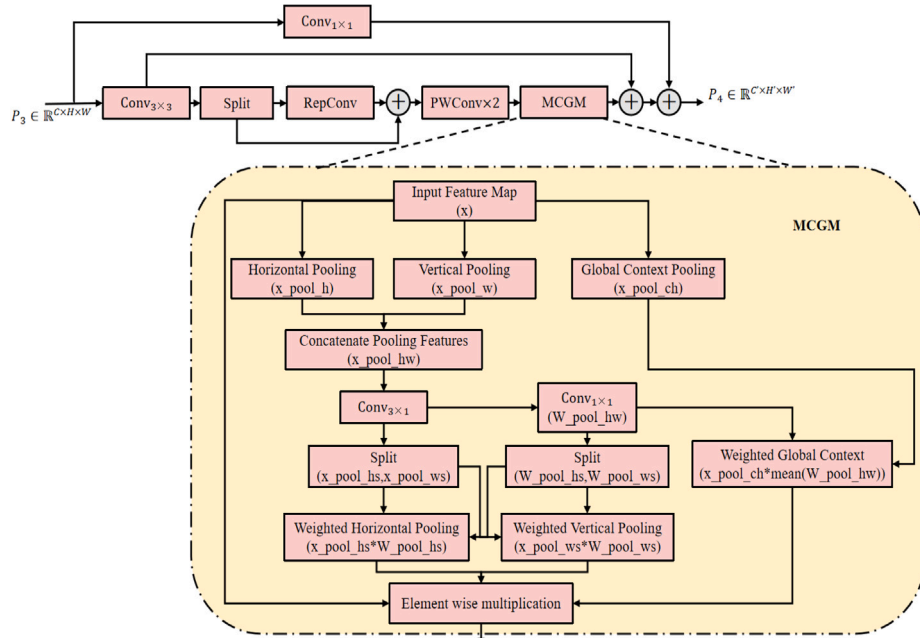


Fig. 4. Structure of EP4MCGM block.

Table 1

Detailed architectures of MPCPKI. Assume input image size = $640 \times 640 \times 3$. To illustrate how these feature maps are generated, consider the $P3$ feature map produced by the BasicBlocks-P3 module (#6). This module consists of two basic blocks, each containing two 3×3 convolutional layers with 128 output channels. The first convolutional layer in each block uses a stride of 2 for downsampling, effectively reducing the spatial dimensions. In contrast, the second convolutional layer employs a stride of 1 to maintain the feature map size. Consequently, the BasicBlocks-P3 module outputs a feature map of size $80 \times 80 \times 128$.

#	Block name	Block nums	Kernel size	Channels	Stride	Padding	Output size
1	ConvNormLayer1	1	3×3	32	2	1	$320 \times 320 \times 32$
2	ConvNormLayer2	1	3×3	32	1	1	$320 \times 320 \times 32$
3	ConvNormLayer3	1	3×3	64	1	1	$320 \times 320 \times 64$
4	MaxPool2d	1	3×3	—	2	1	$160 \times 160 \times 64$
5	BasicBlocks-P2	2	$\begin{bmatrix} 3 \times 3 \\ 3 \times 3 \end{bmatrix} \times 2$	$\begin{bmatrix} 64 \\ 64 \end{bmatrix} \times 2$	1	1	$160 \times 160 \times 64$
6	BasicBlocks-P3	2	$\begin{bmatrix} 3 \times 3 \\ 3 \times 3 \end{bmatrix} \times 2$	$\begin{bmatrix} 128 \\ 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	1	$80 \times 80 \times 128$
7	EP4MCGM	2	—	—	—	—	$40 \times 40 \times 256$
8	CSPPKINetP5	2	—	—	—	—	$20 \times 20 \times 512$

(MCGM), adaptively enhances the spatial features for the $P4$ feature layer. The multipath cooperative mechanism employs directional pooling (horizontal and vertical) and global context aggregation in shallow layers, thereby capturing subtle structural variations. In addition, its gating mechanism suppresses redundant features and emphasizes important regions, enhancing fine-grained detail detection. The feature transformation process is defined as follows.

$$P4 = m(P3) + \text{Conv}_{3 \times 3}(P3) + \text{Conv}_{1 \times 1}(P3) \quad (1)$$

where $m(P3)$ represents the primary transformation process that selectively enhances informative feature representations while suppressing irrelevant information. It is defined as

$$m(P3) = \text{MCGM} \left(\text{PWConv} \left(\text{RepConv} \left(\text{Split}_{\frac{1}{4}} \left(\text{Conv}_{3 \times 3}(P3) \right) \right) + \text{Split}_{\frac{3}{4}} \left(\text{Conv}_{3 \times 3}(P3) \right) \right) \right) \quad (2)$$

Here, PWConv denotes a pointwise convolution that fuses channel-wise features. In contrast, RepConv (Ding et al., 2021) is a reparameterized convolution block that improves computational efficiency by restructuring convolutions into lightweight operations. These components work together to refine the extracted features before they are processed by the MCGM.

Before being processed by MCGM, the input feature map undergoes convolutional transformations and splitting operations to generate x , which serves as the input to MCGM:

$$x = \text{PWConv} \left(\text{RepConv} \left(\text{Split}_{\frac{1}{4}} \left(\text{Conv}_{3 \times 3}(P3) \right) \right) + \text{Split}_{\frac{3}{4}} \left(\text{Conv}_{3 \times 3}(P3) \right) \right) \quad (3)$$

MCGM takes x as input and refines its feature representations using multi-directional pooling and adaptive weighting mechanisms. MCGM was introduced to enable efficient multiscale feature refinement, which enables selective feature enhancement while reducing redundancy in deep feature processing. The final output of MCGM is computed as

$$\text{MCGM}(x) = x \cdot \sigma(x_{\text{pool}_{hs_w}}) \cdot \sigma(x_{\text{pool}_{ws_w}}) \cdot \sigma(x_{\text{pool}_{ch_w}}) \quad (4)$$

where $\sigma(\cdot)$ denotes the sigmoid activation function and ensures adaptive scaling of the feature maps. The terms $x_{\text{pool}_{hs_w}}$ and $x_{\text{pool}_{ws_w}}$ correspond to the horizontally and vertically pooled features weighted by their respective coefficients, and $x_{\text{pool}_{ch_w}}$ represents the globally pooled feature map with context-aware adjustments.

To compute these weighted feature representations, MCGM first applies directional pooling to extract horizontal and vertical feature statistics $x_{\text{pool}_h} = \text{Pool}_h(x) \in \mathbb{R}^{B \times C \times H \times 1}$ and $x_{\text{pool}_w} = \text{Pool}_w(x) \in \mathbb{R}^{B \times C \times 1 \times W}$. Additionally, a global context representation is derived by applying global average pooling, followed by a convolutional layer $x_{\text{pool}_{ch}} = \text{Conv}(\text{AdaptiveAvgPool2d}(x)) \in \mathbb{R}^{B \times C \times 1 \times 1}$. The concatenated horizontal and vertical features are processed by a convolutional layer

to obtain $x_{\text{pool}_{hw}} = \text{Concat}(x_{\text{pool}_h}, x_{\text{pool}_w}) \in \mathbb{R}^{B \times C \times (H+W) \times 1}$, which is further split into directional components, $x_{\text{pool}_{hs}}$ and $x_{\text{pool}_{ws}}$. Subsequently, an additional convolution operation generates a weighting map $W_{\text{pool}_{hw}} = \text{Conv}(\text{Conv}(x_{\text{pool}_{hw}})) \in \mathbb{R}^{B \times C \times (H+W) \times 1}$. This weight map is processed through a sigmoid activation function $W_{\text{pool}_{hw_AF}} = \sigma(W_{\text{pool}_{hw}}) \in \mathbb{R}^{B \times C \times (H+W) \times 1}$, which is then split into horizontal and vertical weighting coefficients, $W_{\text{pool}_{hs}}$ and $W_{\text{pool}_{ws}}$ respectively. The final weighted feature representations were computed as follows:

$$x_{\text{pool}_{hs_w}} = x_{\text{pool}_{hs}} \cdot W_{\text{pool}_{hs}} \in \mathbb{R}^{B \times C \times H \times 1} \quad (5)$$

$$x_{\text{pool}_{ws_w}} = x_{\text{pool}_{ws}} \cdot W_{\text{pool}_{ws}} \in \mathbb{R}^{B \times C \times 1 \times W} \quad (6)$$

Additionally, the global context feature is adjusted using spatially averaged weighting coefficients:

$$x_{\text{pool}_{ch_w}} = x_{\text{pool}_{ch}} \cdot \frac{\sum_{h=1}^H W_{\text{pool}_{hw_AF}}(h, :)}{H} \quad (7)$$

Integrating MCGM into EP4MCGM enhances the spatial feature selection by capturing structural variations through multipath pooling and adaptive weighting. MCGM emphasizes informative regions while suppressing redundancy and refining the $P4$ feature map for subsequent processing.

As shown in Table 1, the CSPPKINetP5 module (#8) integrates the cross-stage partial (CSP) structure with the PKI and context anchor attention (CAA) modules within the PKI Block (Cai et al., 2024). The CSP strategy splits the input feature map into two branches: one extracts high-level features through intensive transformations, while the other retains spatial information via a direct pathway. This design minimizes redundant computations by processing only half of the feature channels intensively, while the other half is preserved with minimal processing. Additionally, CSP improves gradient propagation, enhancing training efficiency and feature integrity. The PKI module employs depthwise convolutions with varying kernel sizes to capture multi-scale features. By aggregating responses from filters of different receptive fields, PKI enables robust detection across objects of varying sizes while significantly reducing computational costs compared to traditional convolutions. Finally, the CAA module dynamically adjusts feature responses based on spatial dependencies. It applies lightweight horizontal and vertical convolutions followed by a sigmoid activation to generate attention weights, enhancing focus on crucial regions and adapting to variations in object positioning and scale.

Given an input feature defined in Eq. (1), $P4 \in \mathbb{R}^{C' \times H' \times W'}$, where H' and W' are $\frac{1}{16}$ of the height and width of the original image, respectively, the module processes the data to produce the final output $P5$, defined by the following equation:

$$P5 = \text{Conv}_{1 \times 1}(\text{FFN}(S_{P5'}) \parallel \text{Conv}_{1 \times 1}(\text{CAA}(S_{P5''}) \odot \text{PKI}(S_{P5''}) + \text{PKI}(S_{P5''}))) \quad (8)$$

Here, $\parallel$ denotes concatenation, and $\odot$ represents element-wise multiplication.

Specifically, to achieve efficient feature extraction, CSPPKINetP5 first applies a 3×3 convolutional layer to P_4 to produce an intermediate feature map, X_{conv} . This map is then split equally along the channel dimension into $S_{P_5'}$ and $S_{P_5''}$, each with dimensions $\mathbb{R}^{\frac{C''}{2} \times H'' \times W''}$, where H'' and W'' are each $\frac{1}{32}$ of the height and width of the original image. The first branch, $S_{P_5'}$, is processed through a lightweight feed-forward network (FFN) to generate the intermediate output $D = \text{FFN}(S_{P_5'})$, which refines the feature representation while maintaining low computational complexity. The second branch, $S_{P_5''}$, undergoes multiscale processing using the PKI and CAA modules. The PKI module employs a set of depthwise convolutions with kernel sizes (3, 5, 7, 9, 11) to enhance multiscale feature extraction. By aggregating responses from filters of different receptive fields, this structure captures spatial features across varying object sizes, thereby improving robustness to scale variations while maintaining computational efficiency. The output of the PKI module, $\text{PKI}(S_{P_5''})$, is modulated by the CAA module, which dynamically adjusts feature responses based on contextual relevance. The resulting attention-weighted feature map was computed as $A_W = \text{CAA}(S_{P_5''}) \odot \text{PKI}(S_{P_5''})$, which is further refined by incorporating residual connections: $B = A_W + \text{PKI}(S_{P_5''})$. Subsequently, a 1×1 convolution is applied to B , producing the intermediate output $C = \text{Conv}_{1 \times 1}(B)$. Finally, the outputs from both branches C and D are concatenated along the channel dimension, forming $E = D \parallel C$, followed by a 1×1 convolution to generate the final output P_5 . By leveraging the cross-stage partial mechanism to optimize feature propagation, the PKI module to extract multiscale representations, and the Context Anchor Attention module to enhance spatial adaptivity, CSPPKINetP5 achieves a superior balance between detection accuracy and computational efficiency, contributing to the overall robustness of the model.

2.2.3. Frequency-domain context feature fusion (FDCFN)

As shown in Fig. 5, the FDCFN consists of the FDB and the CFFB. This design leverages complementary spatial- and frequency-domain representations to enhance feature extraction, addressing the limitations of conventional spatial-domain methods in handling multiscale variations and long-range dependencies. FDB handles intra-scale feature interactions and effectively preserves fine-grained details through frequency-domain processing. Meanwhile, the CFFB facilitates cross-scale feature fusion and captures long-range dependencies. By integrating these two branches, FDCFN retains high-frequency details while leveraging global contextual cues, significantly improving detection robustness in multiscale and occluded environments.

The FDB enhances the P_5 feature map by incorporating frequency-domain information. Traditional feature extraction relies on spatial-domain operations such as convolution, which struggle with long-range dependencies and subtle texture variations. To address this issue, FDB employs the HiLo (Pan et al., 2022) attention mechanism to decompose feature maps into high-frequency (Hi-Fi) and low-frequency (Lo-Fi) components. Unlike Fourier or wavelet transforms, which rely on pre-defined basis functions, HiLo adaptively partitions the feature space using learnable attention mechanisms, enabling a more flexible and data-driven frequency decomposition. Specifically, P_5 serves as the value representation (V), whereas the query (Q) and key (K) vectors incorporate the positional encodings $Q = K = P_5 + \text{Pos}$ and $V = P_5$. The Hi-Fi component is extracted via local self-attention within non-overlapping windows, capturing fine-grained details. In contrast, the Lo-Fi component is obtained through downsampled self-attention, where average pooling reduces the resolution before applying attention, facilitating long-range dependency modeling. This combination enables FDB to effectively balance fine-grained texture representation and global semantic comprehension, leading to a more robust and discriminative feature space. The outputs from both components were concatenated along the channel dimension, preserving both the

fine-scale texture and large-scale semantic structures. The final representation is refined through a feed-forward network (FFN) and layer normalization (LN) as follows:

$$F5 = \text{LN} [\text{FFN} (\text{LN} [\text{Linear} (\text{HiLo} (Q, K, V) + P_5)]) + \text{LN} [\text{Linear} (\text{HiLo} (Q, K, V) + P_5)]] \quad (9)$$

In the CFFB, multiscale feature maps—namely P_3 , P_4 , and $F5$ —are processed to effectively integrate both local and global contextual information from both the spatial and frequency domains. While FDB enhances feature expressiveness by incorporating frequency information, CFFB ensures the effective fusion of multiscale features, maintaining spatial coherence and structural integrity. Each feature map X in $\text{set}\{P_3, P_4, F5\}$ is first passed through the CAA module to enhance its spatial contextual information its spatial contextual information. The CAA module employs horizontal and vertical convolutions to capture long-range dependencies in both spatial dimensions, followed by sigmoid activation to generate the attention weights. This design choice allows the model to efficiently encode spatial relationships across multiple scales while mitigating information loss due to downsampling. This process is mathematically represented as follows:

$$X' = \text{conv}_{1 \times 1}(\text{CAA}(X)) \quad \text{for } X \in \{P_3, P_4, F5\} \quad (10)$$

For multiscale feature fusion, the frequency-domain enhanced feature map $F5'$ undergoes a deconvolution operation and is processed again by the CAA module to obtain $F5'_{\text{processed}} = \text{CAA}(\text{Deconv}(F5'))$. This processed $F5'_{\text{processed}}$ is then multiplied element-wise by $P4'$, and the result is added to the deconvolved $F5'$ to produce the fused feature map $P4'_{\text{final}} = P4' \odot F5'_{\text{processed}} + \text{Deconv}(F5')$. Similarly, $F5'$ undergoes two consecutive deconvolution operations followed by the CAA module, yielding $F5''_{\text{processed}} = \text{CAA}(\text{Deconv}(\text{Deconv}(F5')))$. This processed $F5''_{\text{processed}}$ is then element-wise multiplied by $P3'$, and the outcome is added to the twice-deconvolved $F5'$ to obtain $P3'_{\text{final}} = P3' \odot F5''_{\text{processed}} + \text{Deconv}(\text{Deconv}(F5'))$. The fusion mechanism, named dual-domain attention fusion (DDAF), combines the frequency-domain information from the FDB with the spatial-domain information from the CAA module. Specifically, the frequency-domain-enhanced feature map $F5'$ is fused with the spatial-domain feature maps $P3'$ and $P4'$ through a combination of deconvolution, attention-based processing, and element-wise multiplication. This allows the network to leverage fine-grained texture details from the frequency domain and long-range spatial dependencies from the spatial domain. The fused feature maps $P4'_{\text{final}}$ and $P3'_{\text{final}}$ are subsequently processed using the RepC3 module as implemented in this work, resulting in

$$P_i'^{\text{RepC3}} = \text{RepC3}(P_i'_{\text{final}}) \quad \text{for } i \in \{3, 4\} \quad (11)$$

Finally, the processed feature maps $P4'^{\text{RepC3}}$, $P3'^{\text{RepC3}}$, and $F5'$ are concatenated to generate the final FDCFN output.

$$\text{FDCFN} = \text{Concat}(P4'^{\text{RepC3}}, P3'^{\text{RepC3}}, F5') \quad (12)$$

2.2.4. Loss function (SIOU)

Background misidentification and missed detections are common in beef cattle detection, particularly in real-world agricultural settings where cattle may be partially occluded by fences or other livestock and appear in various orientations due to movement. To address these issues, we incorporate SCYLLA-IoU (SIOU) loss into the detection framework. Unlike traditional intersection-over-union (IoU)-based losses, SIOU integrates the angle between the predicted and ground-truth bounding boxes, improving the model's ability to handle variations in object orientation that are common in farm environments. This angular consideration reduces background misidentification and improves foreground-background separation. Specifically, the RTDETR loss function, initially consisting of bounding box regression loss and classification loss, was modified by redefining the bounding box regression loss as a weighted sum of SIOU (Gevorgyan, 2022) and L1 loss, replacing the previous combination of generalized intersection over

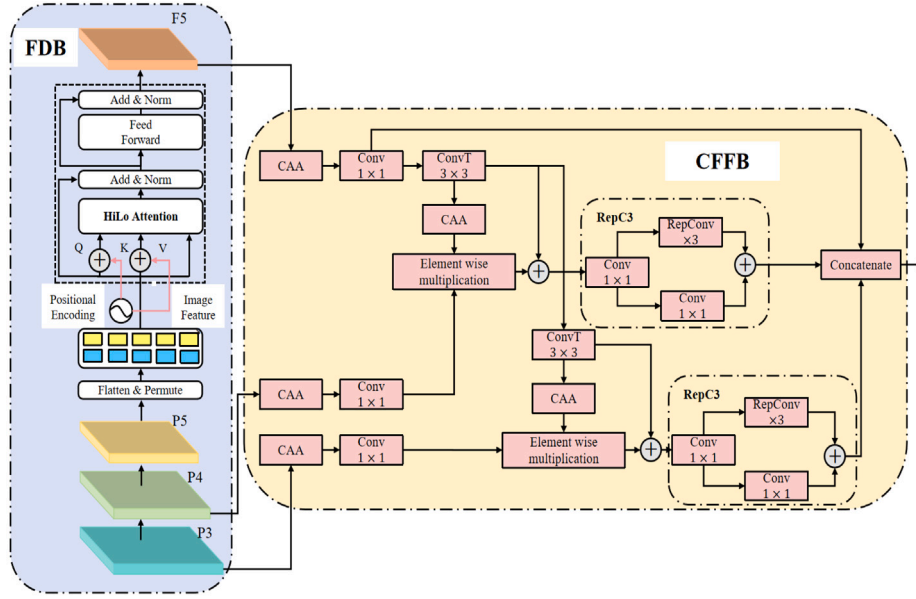


Fig. 5. Structure of the FDCFN block.

union (GIoU) (Rezatofighi et al., 2019) and L1 loss. The classification loss remains varifocal loss (VFL) (Zhang et al., 2021). By introducing angle loss through SIoU, the bounding-box regression accuracy improved, enhancing the overall detection performance. The model is optimized during training by minimizing both the bounding box regression loss and the classification loss, as defined below:

$$L_{cls} = \begin{cases} -q(q \log(p) + (1-q) \log(1-p)), & \text{if } q > 0 \\ -\alpha p^k \log(1-p), & \text{if } q = 0 \end{cases} \quad (13)$$

where p represents the predicted probability of a positive sample and $1-p$ denotes the predicted probability of a negative sample. q distinguishes between the foreground ($q > 0$) and background ($q = 0$). The hyperparameter α mitigates the class imbalance between positive and negative samples by adjusting the weighting of the loss function, whereas k addresses the imbalance between easy and hard samples by modulating the focus on different sample difficulties. In our experiments, we adopted the default values for α and k as specified in VarifocalNet (Zhang et al., 2021) to ensure consistency with established methodologies.

$$L_{box}(\hat{b}, b) = \lambda L_{SIoU}(\hat{b}, b) + \lambda_{L1} \|\hat{b} - b\|_1 \quad (14)$$

Here, $\hat{b}$ represents the predicted bounding box, b is the actual bounding box, and L_{SIoU} is the SIoU loss. The details of its definition can be found in Gevorgyan (2022). The weight parameters λ and λ_{L1} were used to adjust the contributions of L_{SIoU} and L1 loss in the overall L_{box} loss, respectively. In our experiments, λ and λ_{L1} are set to their default values (Zhao et al., 2024a).

2.3. Evaluation indicators

Unlike classification tasks, detection models output unstructured data, and the number, location, and size of the detected objects are unknown in advance. Thus, the intersection over union (IoU) (Rezatofighi et al., 2019) is used to quantify the overlap between the predicted box (A) and the ground-truth box (B). A detection is considered correct if the IoU exceeds a threshold. α . Otherwise, it is incorrect. IoU is calculated as follows:

$$IoU = \frac{S_{A \cap B}}{S_{A \cup B}} \quad (15)$$

where $S_{A \cap B}$ is the area of overlap and $S_{A \cup B}$ is the total area covered by both boxes.

In object detection, targets are the objects of interest, and everything else is background. Bounding box predictions are categorized into four types: true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN). TP refers to correctly detected targets, while FP refers to background misidentified as targets ($IoU < \alpha$). FN refers to missed targets, and TN refers to the correctly predicted background. Based on this, Precision and Recall were calculated as follows:

$$Precision = \frac{TP}{TP + FP} \times 100\% \quad (16)$$

$$Recall = \frac{TP}{TP + FN} \times 100\% \quad (17)$$

Precision measures the proportion of correctly predicted positives, while recall measures the proportion of true positives detected. Ideally, both Precision and Recall should be high, but they are inversely related—improving one may reduce the other. To balance both, average precision (AP) is used. AP@0.5 uses a lenient IoU threshold, while AP@0.5–0.95 is stricter, requiring high detection accuracy across multiple IoU thresholds. Higher AP@0.5 and AP@0.5–0.95 indicate robust performance across different IoU thresholds.

Beyond accuracy, parameters such as the number of parameters (Params), GFLOPs, inference time, and FPS are crucial for evaluating the overall performance. Fewer Params and GFLOPs indicate the model is lighter and has lower computational requirements, making it more suitable for edge devices. A shorter inference time and higher FPS indicate faster processing, making the model more appropriate for real-time applications.

3. Results and discussion

3.1. Experimental setup

In this study, the proposed EFSF-RBCD and other comparison methods were implemented using Python 3.9.18 and PyTorch 1.13.1. The experiments were conducted on a computer equipped with an Ubuntu 20.04.6 operating system, two GeForce GTX 3090 GPUs, an Intel® Xeon® Gold 6226R CPU @ 2.90 GHz processor, 257563MB of memory, and a 16TB hard drive. In the proposed EFSF-RBCD method, the input images were preprocessed for model training. This includes resizing images ($640 \times 640 \times 3$), normalizing the pixel values, and performing data augmentation (Blur, MedianBlur, ToGray, CLAHE), among other steps. These steps aim to enhance the robustness and generalization

Table 2

Comparison with other state-of-the-art object detection methods based on the beef cattle object detection dataset.

Models	Backbone	AP0.5↑	AP0.5-0.95↑	Params↓	Gflops↓	FPS bs=1↑
YOLOv3 (Redmon and Farhadi, 2018)	–	87.8%	55.4%	117.7M	154.5	130.9
YOLOv5-l (Jocher, 2020)	–	87.7%	55.6%	88.4M	107.6	116.3
YOLOv6-l (Li et al., 2022)	–	86.7%	56.6%	59.5M	150.5	125.6
YOLOX-x (Ge et al., 2021)	–	87.4%	58.6%	98.9M	141.0	42.6
YOLOv8-l (Jocher et al., 2023)	–	87.9%	57.4%	83.6M	164.8	92.5
YOLOv10-l (Wang et al., 2024b)	–	89.3%	57.5%	49.8M	120.0	108.8
YOLOv11-l (Jocher and Qiu, 2024)	–	88.9%	58.3%	48.8M	86.6	89.2
Libra R-CNN (Pang et al., 2019)	ResNet18	83.8%	51.1%	28.3M	69.8	50.0
Cascade R-CNN (Cai and Vasconcelos, 2019)	ResNet18	86.2%	56.5%	56.3M	97.9	34.4
Dynamic R-CNN (Zhang et al., 2020a)	ResNet18	85.0%	58.9%	28.3M	69.8	49.5
Double-Head R-CNN (Wu et al., 2020)	ResNet18	86.6%	52.6%	33.9M	344.0	15.9
DINO (Zhang et al., 2022)	ResNet18	88.5%	54.3%	32.8M	102	22.8
H-DINO (Jia et al., 2023)	ResNet18	86.0%	53.4%	32.7M	120	19.2
DAB-DETR (Liu et al., 2022)	ResNet18	83.4%	45.2%	31.0M	24.2	32.6
Align-DETR (Cai et al., 2023)	ResNet18	85.6%	53.9%	32.8M	102	28.8
Deformable DETR (Zhu et al., 2021)	ResNet18	86.6%	55.8%	23.5M	59.5	32.6
VarifocalNet (Zhang et al., 2021)	ResNet50	88.7%	59.2%	32.7M	75.5	30.7
ATSS (Zhang et al., 2020b)	ResNet50	85.5%	50.5%	32.1M	80.5	38.2
DDOD (Chen et al., 2021)	ResNet50	88.5%	58.9%	32.2M	71.1	34.2
TOOD (Feng et al., 2021)	ResNet50	87.2%	58.2%	32.0M	78.8	28.2
RTDETR (Zhao et al., 2024a)	ResNet18	88.3%	58.3%	38.6M	56.9	106.1
EFSF-RBCD (Ours)	MPCPKI	90.3%	59.6%	26.4M	50.8	100.3

capabilities of the model. The batch size was set to 16, and we trained for 50 epochs. We used the AdamW optimizer to train the detector with a learning rate of 0.0001, a momentum of 0.9, and a weight decay parameter of 0.0001.

3.2. Comparisons with the state-of-the-art object detection methods

We evaluated the performance of the proposed EFSF-RBCD method on the Beef-Cattle dataset and compared it with several classic and state-of-the-art object detection methods. The results are summarized in Table 2. To ensure a fair comparison, we standardized the input size to $(3 \times 640 \times 640)$ when testing the GFLOPs and FPS metrics, as the input shape influenced these metrics. FPS was computed from the inference time of the model.

The EFSF-RBCD method achieved an AP@0.5 of 90.3% and an AP@0.5–0.95 of 59.6%, with 26.4M parameters, a computational cost of 50.8 GFLOPs, and a processing speed of 100.3 FPS. In terms of AP@0.5 and AP@0.5–0.95, EFSF-RBCD outperformed all the object detection methods listed in Table 2, including the highly competitive and widely recognized YOLOv10-l and YOLOv11-l models. Specifically, EFSF-RBCD improved AP@0.5 by 1.0% and 1.4% compared to YOLOv10-l and YOLOv11-l, respectively, and enhanced AP@0.5–0.95 by 2.1% and 1.3%, respectively. Additionally, EFSF-RBCD reduced the number of parameters by 46.9% and 45.9% and the computational cost by 57.6% and 41.3% compared to YOLOv10-l and YOLOv11-l, respectively. Compared to the baseline model RTDETR, EFSF-RBCD improved AP@0.5 and AP@0.5–0.95 by 2.0% and 1.3%, respectively, while simultaneously reducing the number of parameters and computational costs by 31.6% and 10.7%, respectively. Compared to the baseline RTDETR, which achieves 106.1 FPS, our proposed model, EFSF-RBCD, achieves 100.3 FPS by integrating the EP4MCGM, CSPPKINetP5, and FDCFN modules. Although our method has a slight decrease in frame rate compared to the baseline RTDETR, it achieves the best performance and is significantly higher than the traditional real-time threshold of 30 FPS (Huang et al., 2008; Luo et al., 2018; Cao et al., 2020; Guo et al., 2022; Yang et al., 2022a). Moreover, our approach substantially improves detection accuracy while reducing computational cost. This effective balance between real-time performance and detection accuracy highlights the potential of EFSF-RBCD for practical applications in resource-constrained environments.

3.3. Visualization of detection performance in complex scenarios

As shown in Fig. 6, real-world beef cattle detection is challenging due to occlusion, scale variation, background clutter, and inconsistent lighting. We selected RTDETR, YOLOv10, and YOLOv11 as comparison methods. RTDETR serves as the baseline and represents cutting-edge transformer-based algorithms; YOLOv10 and YOLOv11 are the latest YOLO versions known for their efficiency and accuracy. Misdetectors or missed detections by these three methods are highlighted with blue-dashed bounding boxes, facilitating a direct comparison with the proposed EFSF-RBCD approach. In Fig. 6(a), RTDETR, YOLOv10, and YOLOv11 struggle to detect small, distant cattle in indoor sheds, whereas the P4 feature layer selection module of EFSF-RBCD enhances fine-detail capture and improves small-target recognition. Fig. 6(b) shows how background clutter leads the first three methods to misidentify farming workers as cattle, but EFSF-RBCD leverages a P5 feature layer selection module—based on the Cross-Stage Partial PKI Network—to reduce false positives. Fig. 6(c) illustrates how under shadowed or low-light conditions, RTDETR, YOLOv10, and YOLOv11 often misclassify cattle as background or fail to detect them altogether. By contrast, the FDCFN of EFSF-RBCD maintains high accuracy in low-light conditions by integrating local and global contexts. Finally, Fig. 6(d) highlights partially occluded cattle in crowded outdoor environments, where advanced feature selection and fusion strategies enable EFSF-RBCD to detect obscured animals that other methods miss robustly. Overall, these visualizations demonstrate that EFSF-RBCD not only addresses small-target detection, background clutter, weak lighting, and occlusion but also exhibits greater robustness and accuracy across diverse real-world scenarios. By integrating advanced feature selection and fusion techniques, it outperforms RTDETR, YOLOv10, and YOLOv11 in terms of detection performance and adaptability, underscoring its potential for broader applications in precision cattle farming.

3.4. Ablation study

3.4.1. Effectiveness of the proposed module

The proposed method comprises four key components: (1) an EP4MCGM-based efficient P4 feature layer selection module, (2) a CSPPKINetP5-based P5 feature layer selection module, (3) an FDCFN, and (4) a novel SIoU bounding box regression loss function. The results presented in Table 3 demonstrate the impact of adding each component on performance.



Fig. 6. Detection comparison in complex scenarios.

Table 3

Experimental results of the effectiveness of the proposed module.

#	Baseline	EP4MCGM	CSPPKINetP5	FDCFN	SIoU	Precision ↑	Recall↑	AP0.5↑	AP0.5-0.95↑	Params↓	Gflops↓
1	✓					85.1%	84.9%	88.3%	58.3%	38.6M	56.9
2	✓	✓				86.8%	85.6%	89.6%	59.0%	38.8M	55.2
3	✓	✓	✓			87.3%	85.5%	89.9%	59.6%	29.2M	51.1
4	✓	✓	✓	✓		88.1%	85.3%	90.3%	59.1%	26.4M	50.8
5	✓	✓	✓	✓	✓	88.5%	85.8%	90.3%	59.6%	26.4M	50.8

The results demonstrate consistent performance enhancements with each added module, with each contributing uniquely to the detection accuracy and computational efficiency, and incorporating the EP4MCGM module (#2 $vs.$ #1) increased precision by 1.7%, recall by 0.7%, AP@0.5 by 1.3%, and AP@0.5–0.95 by 0.7%, while reducing GFLOPs, indicating more efficient utilization of the P4 feature layer. These substantial performance gains justify the addition of this module. Further enhancements were achieved using the CSPPKINetP5 module (#3 $vs.$ #2), increasing the precision by 0.5%, AP@0.5 by 0.3%, and AP@0.5–0.95 by 0.6%. Additionally, this module reduces parameters by 9.6M and GFLOPs by 4.1, significantly improving efficiency while maintaining high recall performance. The FDCFN module (#4 $vs.$ #3) further improves precision by 0.8% and AP@0.5 by 0.4% while maintaining a strong performance in AP@0.5–0.95. It also reduces parameters by 2.8M and GFLOPs by 0.3, further optimizing the efficiency. Finally, the full EFSF-RBCD model (#5 $vs.$ #4), incorporating the SiOU module, improved precision (0.4%), recall 0.5%, and AP@0.5–0.95 (0.2%). The parameters and GFLOPs remained stable at 26.4M and 50.8, respectively. These results highlight that EP4MCGM contributes the most significant gains in accuracy, particularly in Precision and AP@0.5, making it a crucial addition for improving overall performance. Although each module addition resulted in incremental improvements, the most substantial gains came from the full integration of all components in Model (#5). Compared to the baseline model RTDETR(#1), EFSF-RBCD(#5) significantly improved in both detection accuracy and computational efficiency, increasing precision by 3.4%, recall of 0.9%, AP@0.5 of 2.0%, and AP@0.5–0.95 of 1.3%. Furthermore, the final model reduced parameters by 12.2 M (from 38.6 to 26.4 M) and GFLOPs by 6.1 (from 56.9 to 50.8 M). These results highlight that combining the EP4MCGM, CSPPKINetP5, FDCFN, and SiOU modules in EFSF-RBCD provides a balanced solution, greatly improving both accuracy and efficiency over the baseline model.

3.4.2. Effectiveness of the feature extraction network

To evaluate the effectiveness of the proposed MPCPKI backbone in beef cattle detection, we compared it with various CNN- and Transformer-based architectures known for their strong performance in general object detection. These include SwinTransformer (Liu et al., 2021b), CSwinTransformer (Dong et al., 2022), VanillaNet (Chen et al., 2023c), FasterNet (Chen et al., 2023b), EfficientViT (Liu et al., 2023), ConvNextV2 (Woo et al., 2023), EfficientFormerV2 (Li et al., 2023), RepViT (Wang et al., 2024a), and UniRepLkNet (Ding et al., 2024). As shown in Table 4, MPCPKI outperformed all other backbones, achieving the highest precision (88.5%), recall (85.8%), AP@0.5 (90.3%), and AP@0.5–0.95 (59.6%). Specifically, compared with the best-performing transformer backbone, SwinTransformer, MPCPKI increased precision by 0.9%, recall by 5.5%, AP@0.5 by 3.0%, and AP@0.5–0.95 by 4.5%. Similarly, compared with the best-performing CNN-based backbone, FasterNet and MPCPKI achieved gains of 3.2% in precision, 5.7% in recall, 3.7% in AP@0.5, and 5.2% in AP@0.5–0.95. These results indicate that the multipath cooperative processing and PKI modules of MPCPKI enhance its ability to capture multiscale and context-aware features, which are crucial for detecting cattle in complex environments. The substantial improvements in Recall and AP@0.5–0.95 further demonstrate its effectiveness in handling challenges such as occlusion, varying lighting conditions, and irregular object shapes commonly found in cattle monitoring scenarios.

3.4.3. Effectiveness of the SiOU loss

To evaluate the effectiveness of SiOU, we compared it with several IoU-based loss functions, including the generalized IoU (GIoU) (Rezatofighi et al., 2019), complete IoU (CIoU) (Zheng et al., 2020), Inner-IoU (Zhang et al., 2023), Shape-IoU (Zhang and Zhang, 2023),

Table 4
Ablation study with different feature extraction network.

Feature extraction network	Precision↑	Recall↑	AP0.5↑	AP0.5-0.95↑
SwinTransformer (Liu et al., 2021b)	87.6%	80.3%	87.3%	55.1%
CSwinTransformer (Dong et al., 2022)	84.6%	78.5%	84.7%	52.4%
VanillaNet (Chen et al., 2023c)	84.8%	78.6%	85.5%	53.7%
FasterNet (Chen et al., 2023b)	85.3%	80.1%	86.6%	54.4%
EfficientViT (Liu et al., 2023)	82.8%	76.0%	81.7%	49.1%
ConvNextV2 (Woo et al., 2023)	84.7%	74.5%	80.5%	48.1%
EfficientFormerV2 (Li et al., 2023)	85.9%	80.2%	85.5%	52.5%
RepViT (Wang et al., 2024a)	86.1%	77.8%	84.6%	51.2%
UniRepLNet (Ding et al., 2024)	85.3%	79.1%	85.4%	52.3%
MPCPKI	88.5%	85.8%	90.3%	59.6%

Table 5
Ablation study with different loss functions.

Loss functions	Precision↑	Recall↑	AP0.5↑	AP0.5-0.95↑
GIoU (Rezatofighi et al., 2019)	88.1%	85.3%	90.3%	59.1%
CIoU (Zheng et al., 2020)	88.1%	85.2%	90.0%	58.7%
Inner-IoU (Zhang et al., 2023)	87.5%	85.0%	90.0%	59.4%
Shape-IoU (Zhang and Zhang, 2023)	87.4%	85.2%	89.8%	59.0%
Powerful-IoU (Liu et al., 2024)	87.5%	85.4%	89.8%	58.7%
SIoU	88.5%	85.8%	90.3%	59.6%

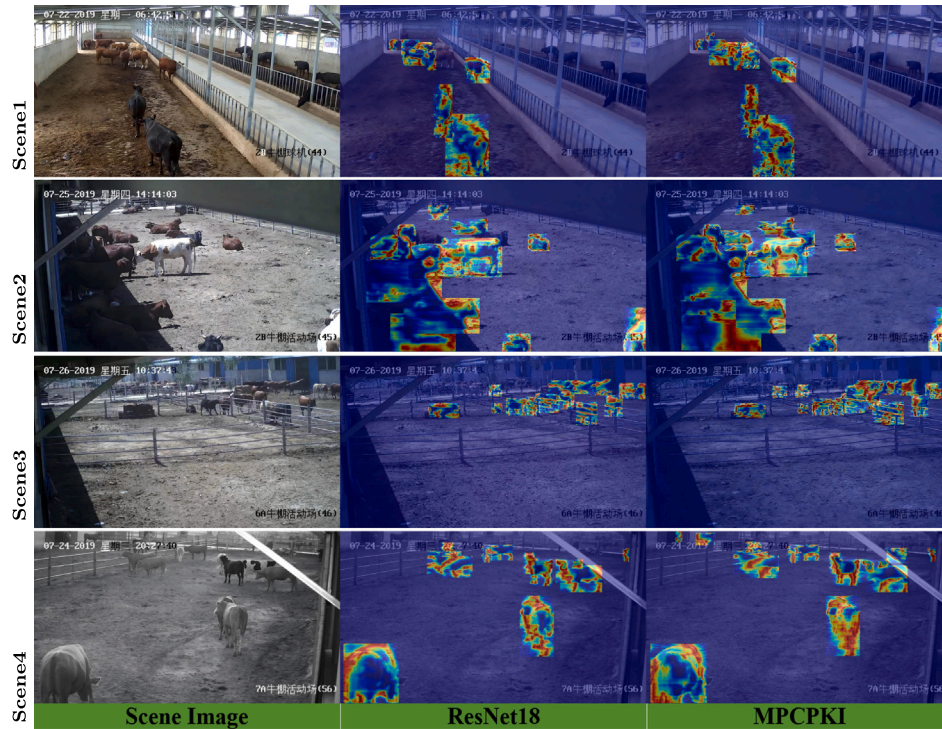


Fig. 7. Comparison of heatmaps. The first column shows original frames from four video scenes. The second and third columns show the heatmaps generated by the feature extraction networks of the RTDETR and EFSF-RBCD models, respectively.

and Powerful-IoU (Liu et al., 2024). As shown in Table 5, SIoU outperformed all other methods across the key performance metrics: achieving the highest precision (88.5%), recall (85.8%), AP@0.5 (90.3%), and AP@0.5–0.95 (59.6%). Specifically, compared to GIoU, SIoU increased precision by 0.4%, recall by 0.5%, and AP@0.5–0.95 by 0.5%. These results demonstrate that SIoU enhances bounding box localization and foreground–background distinction, making it a better choice for bounding box regression in complex cattle detection scenarios.

3.4.4. Feature extraction network heat map visualization analysis

As shown in Fig. 7, we analyze heat map visualizations generated by the feature extraction network to evaluate how effectively the model focuses on key features of beef cattle in various environmental scenarios. Heat maps were generated using Grad-CAM++ (Chattopadhyay

et al., 2018), an advanced method that enhances the original Grad-CAM technique by improving localization and finer object detection, particularly in complex cases with multiple objects or fine details. In Grad-CAM++, red regions indicate the highest attention, while the yellow regions represent moderate attention, highlighting the focus areas of the model.

To assess the performance of the proposed MPCPKI network, we compared heat maps generated by ResNet-18 and MPCPKI across four cattle farm scenarios. With enhancements from the P4 feature layer selection module (EP4MCGM) and P5 module (CSPPKINetP5), MPCPKI exhibits a more precise focus on regions of interest. It produces more accurate heat maps for detecting small targets, recognizing objects in low-light conditions, and handling occlusions. Our analysis indicated that MPCPKI significantly outperformed ResNet-18 in heat map

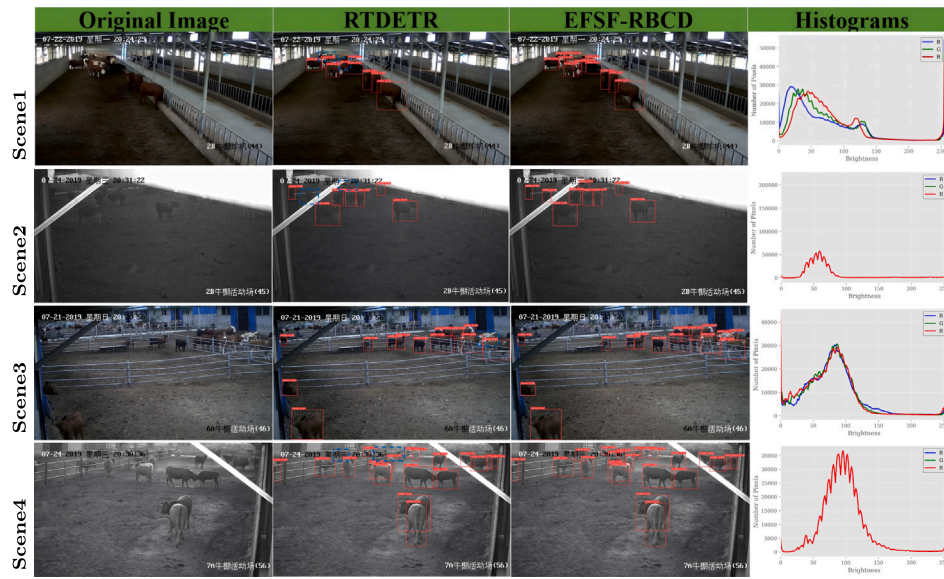


Fig. 8. Detection results under low-light conditions using the EFSF-RBCD method. The first column shows the original frames from four video scenes. The second and third columns display the detections by the RTDETR and EFSF-RBCD models, respectively. The images on the far right show the color histograms for each frame. For convenient understanding, the areas that are difficult for the baseline model to detect are marked with blue dashed boxes.

accuracy, particularly in challenging environments. The fine-grained attention of MPCPKI enables more precise detection of beef cattle in small, occluded, or low-light areas due to the effective integration of feature selection modules that optimize performance while reducing the computational load.

3.4.5. Robustness analysis under low-light conditions

As shown in Fig. 8, we analyze detection results from four different scenes under low-light conditions, each paired with its corresponding color histogram. These histograms illustrate the pixel brightness distribution across RGB channels, offering insights into the lighting conditions. In Scenario 1, the histogram peaks between brightness values of 25 and 50 across all RGB channels, indicating low light. The RTDETR method fails to detect distant or partially occluded cattle, misclassifying them as background. In Scene 2, the red channel histogram peaks at a brightness of approximately 60, with pixel concentrations near 50,000, confirming dim lighting. The RTDETR model struggles as the dark color of the cattle blends with the background, causing misclassification. In Scene 3, the histogram shows a wider distribution across RGB channels, peaking at a brightness of 80 with pixel counts ranging from 10,000 to 30,000, indicating low light. Here, RTDETR misclassifies some cattle as background due to occlusion. In Scene 4, the red channel dominates, peaking at a brightness of 100 with pixel concentrations of approximately 35,000, indicating low light. The RTDETR method misclassifies some cattle occluded by a fence as background. In contrast, the EFSF-RBCD model showed superior detection performance under low-light conditions, as observed in the detection results. The P4 and P5 feature layer selection modules and FDCFN collaboratively enhanced the ability of the model to leverage both local and global contexts, providing a more robust and effective solution for real-time cattle detection in low-light environments.

3.4.6. Robustness analysis in the case of occlusion

As shown in Fig. 9, we analyze the detection results of the EFSF-RBCD method compared to the baseline RTDETR method under different occlusion scenarios. In Scenes 1 and 2, mutual occlusions occurred between the cattle, where one animal was partially hidden by the other. The RTDETR method failed to detect these cattle, misclassifying them as background, due to its limited ability to handle occlusions, especially when multiple cattle were close together and their features blended. In contrast, the EFSF-RBCD method successfully detected cattle by

leveraging the P4 feature layer selection module and multipath cooperative gating mechanism (EP4MCGM) to capture fine-grained details, even under partial occlusion. Scenarios 3 and 4 included both mutual occlusions and those caused by fences. The RTDETR method struggles to detect cattle obstructed by fences because it cannot distinguish occluded objects from the background. However, the EFSF-RBCD method effectively detects cattle, aided by the P5 feature layer selection module (CSPKINetP5), which optimizes feature extraction while minimizing computational load, and the FDCFN, which enhances contextual awareness. The robustness of EFSF-RBCD under occlusion conditions stems from the combination of the P4 and P5 feature layer selection modules, which improve fine detail extraction, and the FDCFN framework, which integrates both local and global contexts. Additionally, the novel SIOU bounding box regression loss function improves localization by incorporating orientation information and enhancing accuracy, even under partial occlusions. In conclusion, EFSF-RBCD significantly outperforms RTDETR in detecting cattle under complex occlusion scenarios, demonstrating its effectiveness for real-world applications where objects are often occluded by environmental factors, such as fences.

3.5. Limitations and follow-up research directions

The experimental results indicate that our method demonstrates promising performance in both accuracy and efficiency. However, detecting cattle at night remains a significant challenge. In extremely dark environments, insufficient lighting severely hampers feature extraction, causing the model to miss cattle that blend into distant, dark backgrounds Fig. 8 (Scene 2). Future work will focus on developing a dedicated dark-adaptation module to improve feature extraction in nighttime videos. Additionally, we plan to incorporate thermal infrared video data to improve detection performance under low-light conditions. Furthermore, a multimodal detection framework that fuses thermal infrared and visible light images we plan to explore to provide complementary information and improve detection robustness in complex nighttime scenarios.

4. Conclusion

In this article, we introduced a novel EFSF-RBCD method for cattle detection, which significantly enhances detection performance. We proposed an MPCPKI feature-extraction network that incorporates the



Fig. 9. Comparison of detection results under occlusion.

EP4MCGM module to integrate low-level features and to improve fine-detail detection. Additionally, it integrates the CSPPKINetP5 module, which efficiently extracts target features from the P5 layer, captures both local and global contextual information, and reduces computational complexity. Furthermore, we introduced a FDCFN that combines a FDB and a CFFB to capture comprehensive contextual information effectively. To further improve detection accuracy, we integrated the SIoU loss function, addressing the limitations of traditional loss functions. Experimental results on the Beef-Cattle dataset demonstrated that EFSF-RBCD achieved an AP@0.5 of 90.3% and an AP@0.5–0.95 of 59.6%, with 26.4M parameters, a computational cost of 50.8 GFLOPs, and a processing speed of 100.3 FPS. Extensive comparisons with state-of-the-art object detection methods verified the effectiveness and robustness of the proposed model. Moreover, these experiments confirmed that our method enables accurate and efficient beef cattle detection, offering technical support for real-time monitoring and precision livestock farming.

CRedit authorship contribution statement

Yang Sun: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yamin Han:** Writing – review & editing, Supervision,

Resources, Funding acquisition. **Xilong Feng:** Visualization, Investigation, Data curation. **Hongming Zhang:** Supervision. **Jie Wu:** Software, Investigation, Data curation. **Yu Zhang:** Data curation. **PeiJie Qin:** Data curation. **Taoping Zhang:** Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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